Arista Access Node YEL-ACC-32 Routing Protocol Flap SOP

Node: YEL-ACC-32 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1795

Description

The YEL-ACC-32 node, an Arista access node operating within the Yelahanka Sub-station site, experiences a Routing Protocol Flap failure domain due to an OSPF adjacency loss caused by a BGP route churn issue. This is a highly dynamic scenario where the network protocol interactions between the BGP and OSPF routing domains result in a temporary instability in the control plane. As the BGP route churn triggers an OSPF adjacency loss, the ISIS LSP flooding mechanism, which is responsible for maintaining the routing table consistency, becomes overwhelmed. This situation necessitates a swift resolution to prevent network instability and ensure uninterrupted traffic flow. The Arista EOS CLI is utilized throughout this SOP to diagnose and rectify the issue.

Pre-requisite Checks

- check: Verify the Yelahanka Sub-station power and environmental conditions
- check: Confirm the redundant link status for YEL-ACC-32
- check: Ensure remote hands are dispatched to the site for assistance

Potential Root Causes

Cause	Probability
BGP route churn	High
OSPF adjacency loss	Medium
ISIS LSP flooding overload	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays the BGP summary information, including the number of peers, routes, and messages exchanged. This helps identify any issues with the BGP protocol.

Command: show ospf database, Expected Output: OSPF database information, Description: This command displays the OSPF database information, including the number of LSAs and their contents. This helps identify any issues with the OSPF protocol.

Command: show isis database, Expected Output: ISIS database information, Description: This command displays the ISIS database information, including the number of LSPs and their contents. This helps identify any issues with the ISIS protocol.
```

Resolution Steps

Step 1: Step: Clear BGP sessions, Command: clear bgp *, Description: This command clears all BGP sessions, which helps resolve the BGP route churn issue. The * indicates that all BGP sessions should be cleared.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf adjacency all, Description: This command resets the OSPF adjacency, which helps resolve the OSPF adjacency loss issue. The all keyword indicates that all OSPF adjacencies should be reset.

Step 3: Step: Clear ISIS LSP flooding, Command: clear isis lsp-flood all, Description: This command clears the ISIS LSP flooding, which helps resolve the ISIS LSP flooding overload issue. The all keyword indicates that all ISIS LSP flooding should be cleared.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information with no issues, Description: This command verifies that the BGP summary information is normal with no issues.

Command: show ospf database, Expected Output: OSPF database information with no issues, Description: This command verifies that the OSPF database information is normal with no issues.

Command: show isis database, Expected Output: ISIS database information with no issues, Description: This command verifies that the ISIS database information is normal with no issues.
```

Roll-back Procedures

- Step: Restore BGP sessions, Command: show run | inc bgp, Description: This command restores the BGP sessions from the running configuration. The inc bgp keyword searches for BGP-related commands in the running configuration.

- Step: Restore OSPF adjacency, Command: show run | inc ospf, Description: This command restores the OSPF adjacency from the running configuration. The inc ospf keyword searches for OSPF-related commands in the running configuration.

- Step: Restore ISIS LSP flooding, Command: show run | inc isis, Description: This command restores the ISIS LSP flooding from the running configuration. The inc isis keyword searches for ISIS-related commands in the running configuration.